Technical Design Document: Async Credit Processing Service

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- **Status:** Implemented
- **Target audience:** Engineering reviewers / tech leads

1. Overview and objectives

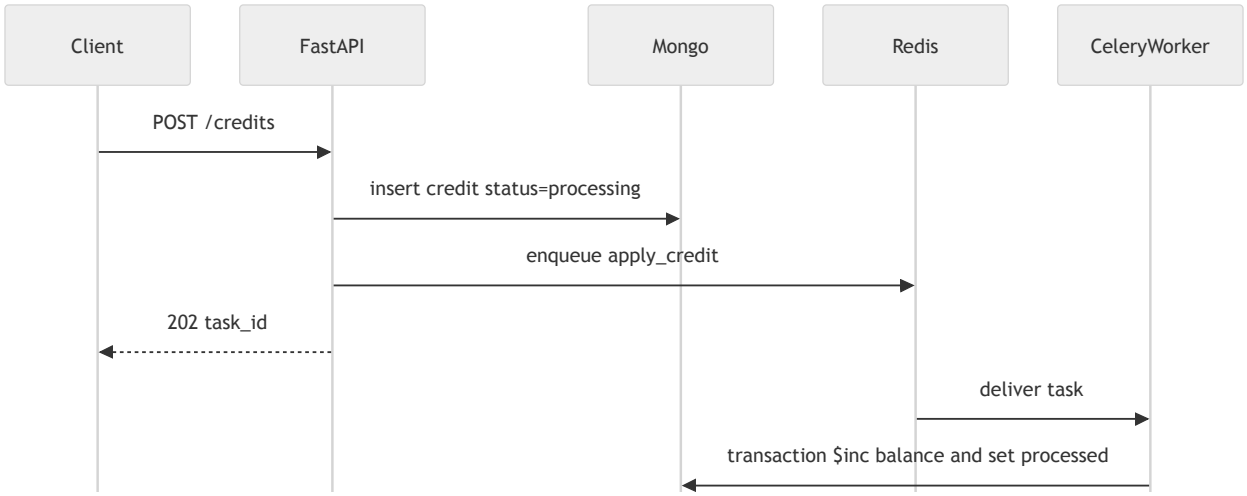
The service accepts a credit over HTTP and updates the user's balance on a worker. The API returns once the credit is stored and queued. A redelivered task must not add the money twice.

- The API returns **202 Accepted** with a `task_id` after the credit is stored as `processing`.
- A Celery worker, with Redis as the broker, executes the credit. Delivery is at-least-once. The balance update is exactly once.
- The transition `processing` → `processed` is committed in one MongoDB transaction with the balance increment.

Clients read MongoDB for the outcome. `GET /credits/{task_id}` returns the credit. `GET /users/{user_id}` returns the committed balance.

2. Architecture and system flow

Four processes share the work. The API and the worker are one Python application, started with different commands from one image.



API. FastAPI on Python 3.11+ and Pydantic. Routes call `CreditApplication` and do not update the balance.

Queue. Redis and Celery, with no result backend. The worker acknowledges late, returns a task if the process dies, and prefetches one message.

Datastore. MongoDB holds `users` and `credits`. Compose runs replica set `rs0` so the balance update can be a multi-document transaction.

3. Key design decisions and rationale

Accept and apply are separate. The request validates, inserts, and enqueues. The balance update does not hold the HTTP call. If Redis rejects the publish, the API marks that credit `failed` and returns **503**.

Status lives on the credit. The balance lives on the user. Two credits for one user can be in flight together. One user document cannot hold both statuses, so the second write would replace the first

and a retry could apply the wrong amount. Each credit is its own document (`processing`, `processed`, or `failed`). The user document stores `user_id` and `balance` only, and is created when the first credit is applied. `failed` means the credit will not be applied.

Redelivery does not double-credit. Celery can deliver a task more than once. Inside the transaction, the worker updates the credit only when its status is still `processing`, then `$inc`s the balance. The worker that loses that update aborts and does not increment. A later delivery of a `processed` or `failed` credit is a no-op.

The idempotency key covers client retries, not worker retries. An optional `Idempotency-Key` header with the same body returns the original task and does not publish again, including when that task `failed`. The same key with a different body returns **409**. A second POST without the header is a second credit.

4. Implementation approach

`POST /credits` accepts `{ "user_id": "u_123", "amount": "150.00" }` and returns **202** `{ "task_id": "<uuid>", "status": "processing" }`. `amount` is a decimal string, greater than zero, at most `1000000.00`, with at most two decimal places. `user_id` is 1–128 characters from letters, digits, `_`, and `-`. Invalid input returns **422**.

User `{ "user_id": "u_123", "balance": "150.00" }`. Credit `{ "task_id": "<uuid>", "user_id": "u_123", "amount": "150.00", "status": "processed" }`. Indexes are a unique `user_id` on users, a unique sparse `idempotency_key` on credits, and `user_id` on credits.

`accept` inserts the processing credit and publishes. A duplicate key returns the original row. `apply` runs the transaction only while the credit is `processing`. The task calls `apply`. Database and network errors retry up to five times with exponential backoff. Any other error, or the last retry, sets `failed` only while the credit is still `processing`. A crash after the insert and before publish can leave a credit `processing` with nothing queued. Nothing sweeps that gap.

5. Testing strategy

Unit tests (`pytest`) build `CreditApplication` on in-memory repositories, a publisher, and a unit of work. They cover one apply, a no-op second apply, idempotent replay and **409**, invalid input, a broker failure, a lost claim, and a database error during apply. A non-retryable worker error is marked `failed` immediately; a first network error stays `processing`.

Integration tests use Compose MongoDB and Redis and write to `credits_test`. They POST through the real repositories, then apply in process, except one test that publishes to Redis.

6. Trade-offs and future work

Redis is the broker because this workload is one task type with retries. RabbitMQ or Kafka would add weight without a routing or retention need here. Clients poll `GET /credits/{task_id}`; a webhook or WebSocket could replace that later. A ledger of immutable entries would make an audit easier, and the conditional credit update would still be required so a redelivery does not append the same entry twice.